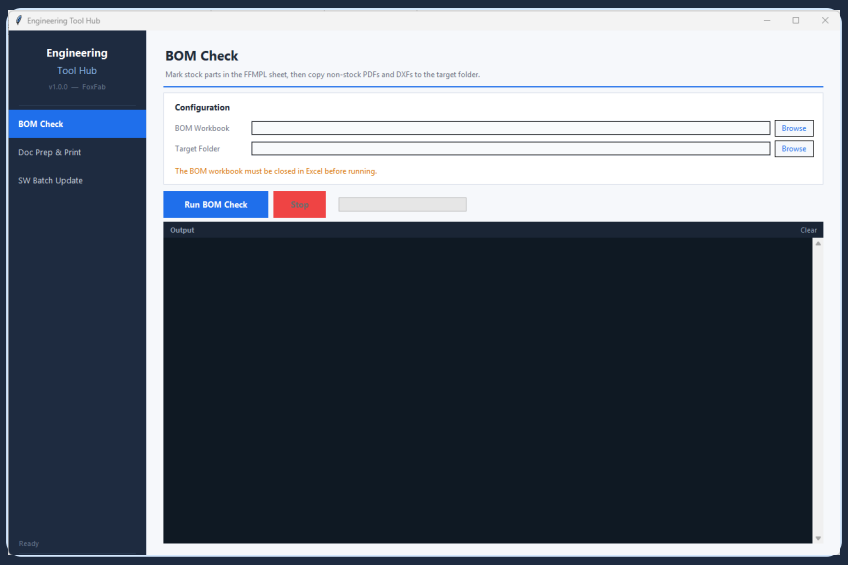
Engineering Tool Hub

User Guide

Version 1.2.0



Lambert Badong

Revision: v1.2.0 Update

FoxFab Internal Documentation

March 2026

Contents

1 Introduction **3**
1.1 Application Layout . . . 3
1.1.1 Terminal colour guide . . . 4

2 Getting Started **5**
2.1 Requirements . . . 5
2.2 Configuring Everything HTTP Server . . . 5
2.3 Launching the Application . . . 6

3 How to Use **9**
3.1 Navigation . . . 9

4 Bom Filler **10**
4.1 What's New in v1.2.0 . . . 10
4.2 Before You Start . . . 10
4.3 Step-by-Step . . . 11
4.4 Understanding the Results . . . 13
4.4.1 Combined-Part Files . . . 13
4.4.2 Punch Program Filter . . . 13
4.5 Stopping Early . . . 13

5 Doc Prep & Print **14**
5.1 Overview — What Gets Printed . . . 14
5.2 Before You Start . . . 14
5.3 Step-by-Step . . . 14
5.4 Preview Options (Optional) . . . 16
5.4.1 Preview FWO . . . 16
5.4.2 Preview BOM . . . 16
5.5 Simulation Mode . . . 17
5.6 Manual Printing . . . 18

6 SW Batch Update **19**
6.1 Step-by-Step . . . 19

7 SW Batch PDF Export **21**
7.1 Step-by-Step . . . 21
7.2 Tips . . . 21

8 File Logger (Parts Tracker) **22**
8.1 Features . . . 22
8.2 Launching . . . 22

9 Training Materials **23**
9.1 Engineering Design Package . . . 23
9.2 Design Reference . . . 23

10 Troubleshooting **24**

11 Quick Reference **25**
11.1 Sidebar Navigation 25
11.2 Button Reference 25
11.3 BOM Column Marking Reference 26
11.4 Print Document Order 26

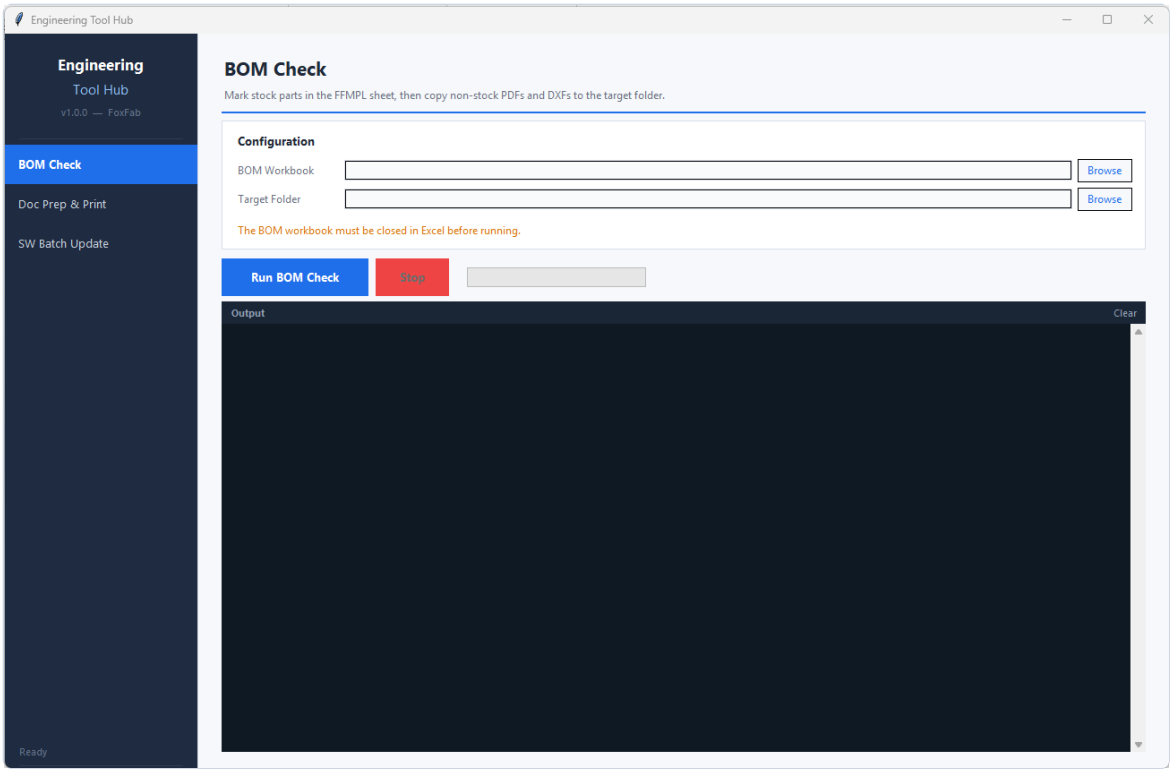
12 Frequently Asked Questions **27**

1 Introduction

Engineering Tool Hub is a FoxFab desktop application that combines seven manufacturing workflow tools into a single interface. Instead of running separate scripts or macros, everything is accessible from one window with a consistent layout.

How to Use	Embedded PDF viewer showing this user guide directly inside the app — no external viewer needed.
Bom Filler	Marks stock parts and copies the correct-revision PDFs and DXFs for non-stock parts into the job folder.
Doc Prep & Print	Builds and prints a complete manufacturing packet — FWO, BOM, CNC files, flats, assemblies — in the correct order.
SW Batch Update	Guides you through running the SolidWorks macro to batch-update custom properties and export DXFs.
SW Batch PDF Export	Step-by-step guide for using SolidWorks Task Scheduler to batch-export drawings as PDFs.
File Logger	Launches the standalone Parts Tracker application for tracking part numbers across job folders.
Training Materials	Batch-prints engineering design reference documents and provides quick access to design tips.

1.1 Application Layout



The window is divided into two main areas:

- **Sidebar (left)** — Click any tool name to switch to it instantly. The active tool is highlighted in blue. The sidebar lists all seven tools in workflow order.
- **Content panel (right)** — Shows the selected tool's config fields, action buttons, progress bar, and terminal output.

> i *The app opens on the **How to Use** tab by default, so new users are immediately oriented before running any tools.*

1.1.1 Terminal colour guide

Colour	Tag	Meaning
Blue	`[INFO]`	General progress update
Green	`[OK]`	Action completed successfully
Amber	`[WARN]`	Something to note, not an error
Red	`[ERROR]`	Something went wrong

2 Getting Started

2.1 Requirements

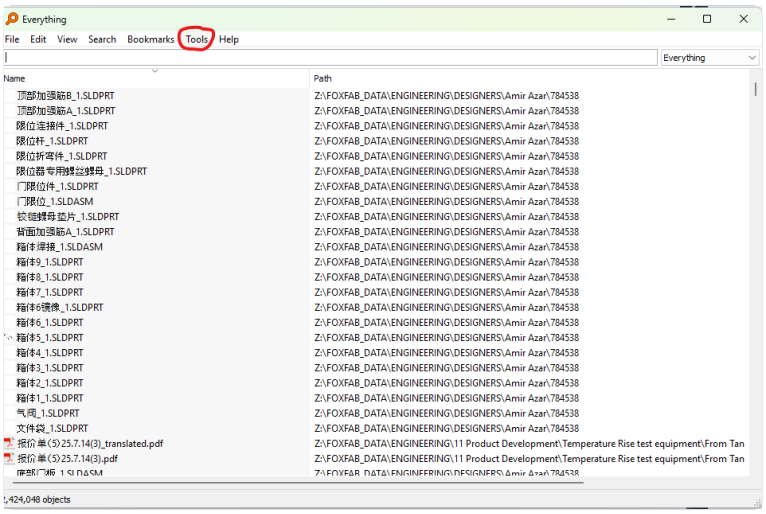
Make sure the following are available on your machine before using the app:

Requirement	Notes
Python (latest)	Download from python.org. During installation, tick **Add Python to PATH**.
Microsoft Excel	Must be installed. Used to read and write BOM spreadsheets.
Adobe Acrobat 2017+	Required for printing. Standard Reader is not sufficient.
Everything (search)	The Everything service must be running and have indexed the `Z:\` drive.
Network drive `Z:\`	Must be mapped and accessible.
FoxFab printer	`\\NPSVR05\FoxFab (Konica Bizhub C360i)` must be installed.

2.2 Configuring Everything HTTP Server

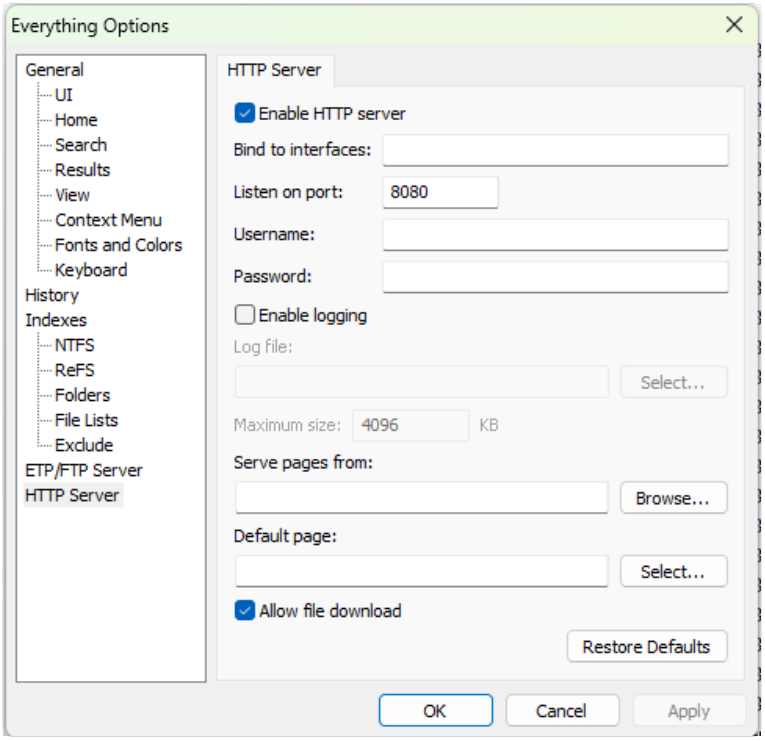
In addition to having the Everything service running, the HTTP server must be enabled for the application to query it.

Step 1. Open the **Everything** application.



Step 2. Go to **Tools** → **Options**.

Step 3. In the left sidebar, select **HTTP Server**.



Step 4. Tick **Enable HTTP server**.

Step 5. In the **Listen port** field, enter `8080`.

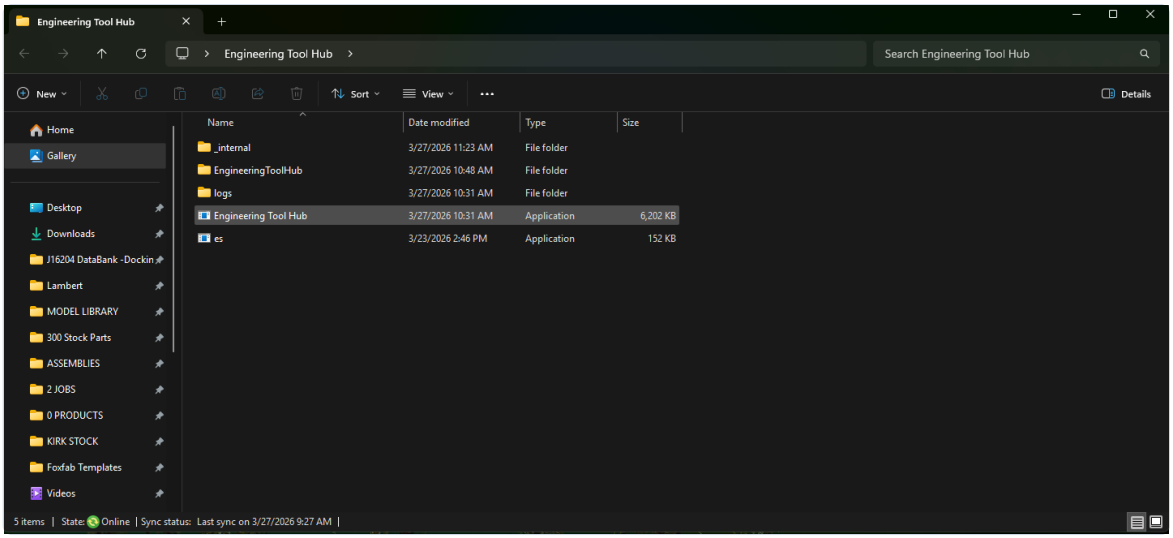
Step 6. Click **OK** to save and close the options window.

! *The HTTP server must be enabled and listening on port `8080` before running Bom Filler or File Logger. If the port is already in use by another application, the Everything search will fail.*

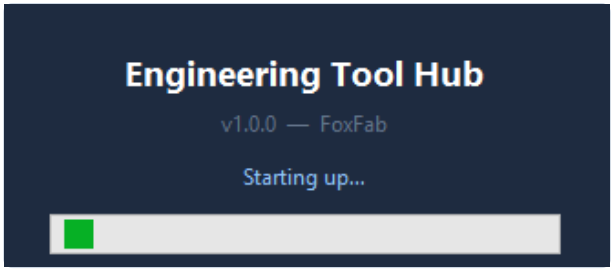
2.3 Launching the Application

Step 7. Navigate to the `Engineering Tool Hub` folder on the shared drive or your desktop.

Step 8. Double-click **Engineering Tool Hub.exe** to launch.

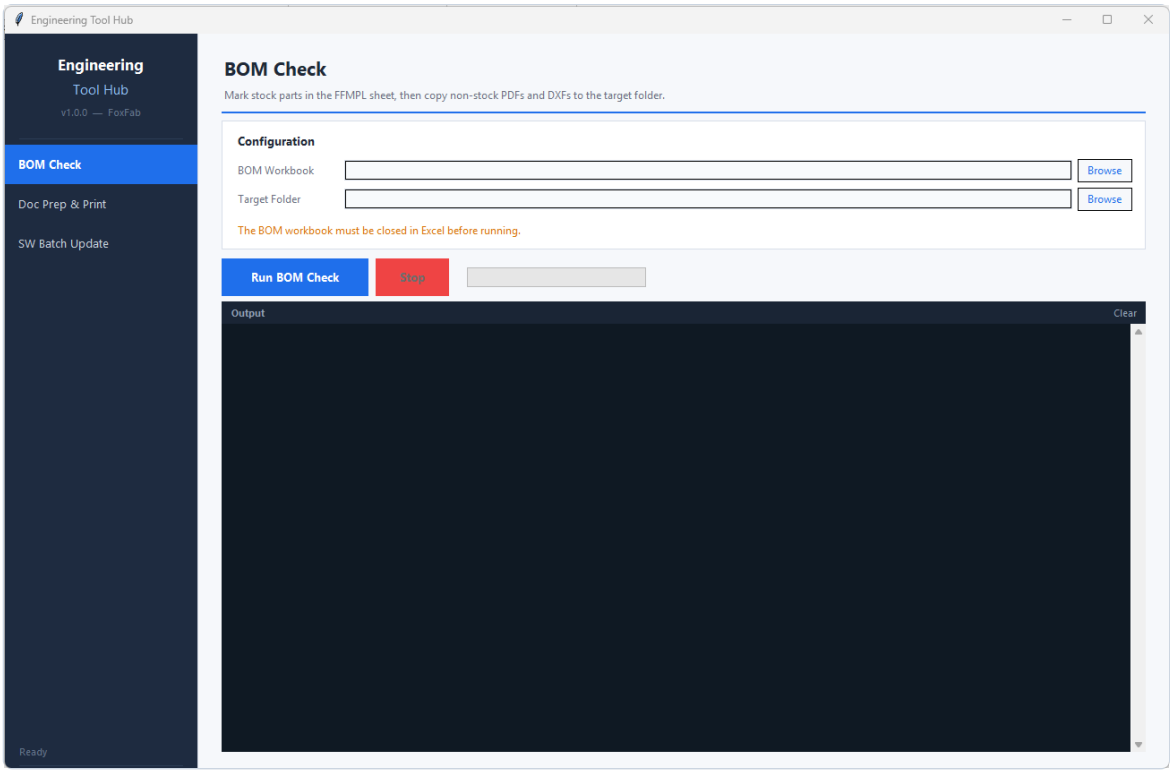


A loading screen appears briefly while the app starts.



! *The entire* `Engineering Tool Hub` *folder is required. Do* ***not*** *move just the* `.exe` *file — it will fail to start without the* `_internal` *folder alongside it.*

Step 9. The main window opens on the **How to Use** tab by default, displaying this user guide in an embedded PDF viewer.


Engineering Tool Hub
Engineering
Tool Hub
v1.0.0 — FoxFab
BOM Check
Doc Prep & Print
SW Batch Update
Ready
BOM Check
Mark stock parts in the FFMPL sheet, then copy non-stock PDFs and DXFs to the target folder.
Configuration
BOM Workbook
Browse
Target Folder
Browse
The BOM workbook must be closed in Excel before running.
Run BOM Check
Stop
Output
Clear

3 How to Use

The **How to Use** tab is the default landing page when the application starts. It renders this user guide (`Engineering_Tool_Hub.pdf`) directly inside the app using an embedded PDF viewer — no external PDF application is required.

3.1 Navigation

- **Prev** / **Next** — Navigate between pages.
- **Page counter** — Displays the current page and total page count (e.g. "Page 5 / 47").
- **Mousewheel** — Scroll vertically within the current page.

> i *If the PDF file is not found alongside the application, the tab will display a message indicating the file is missing. Place `Engineering_Tool_Hub.pdf` next to the `.exe` to restore it.*

4 Bom Filler

Bom Filler reads the **FFMPL** sheet from a job BOM workbook and runs two passes:

1. **Pass 1 — Stock Parts Check.** Each part number is searched against the Stock Parts folder. Matching parts are marked in the spreadsheet.
2. **Pass 2 — Non-Stock File Copy.** For every non-stock part, the highest-revision PDF and DXF are found and copied to your target folder.

4.1 What's New in v1.2.0

- **Auto-populate Target Folder** — When you select a BOM file from a `204 BOM\` directory, the target folder is automatically filled with the sibling `202 PDFs_Flats` path. You can accept or change it.
- **Search Result Caching** — Everything CLI results are cached per run, eliminating redundant lookups for repeated part numbers. Large BOMs run significantly faster.
- **Stock Index Pre-fetch** — The entire stock folder is indexed in a single search call at run start instead of per-part lookups. Pass 1 is dramatically faster for large BOMs.
- **Punch Program Filter** — PDFs and DXFs under `PUNCH PROGRAMS` paths are automatically skipped, preventing punch-specific file variants from being used for general manufacturing.
- **Flexibar Part Support** — Parts with the `245-` prefix are recognized as flexibar parts that never have DXF files. The DXF column is marked `N/A` instead of searching, eliminating false "not found" warnings.
- **Combined-Part File Support** — Detects combined-part filenames like `240-90123_124.pdf` and automatically cross-marks all covered part rows in the BOM.
- **Configuration Variant Handling** — Stock check handles config-variant parts (e.g. `250-30834_001`) by stripping the `_###` suffix and retrying with the base number.

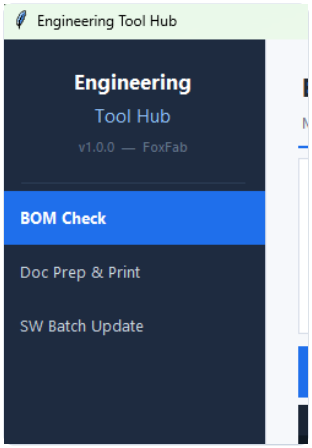
4.2 Before You Start

> ! *The BOM Excel file **must be closed** before running Bom Filler. If it is open in Excel, the tool cannot write to it and will report an error.*

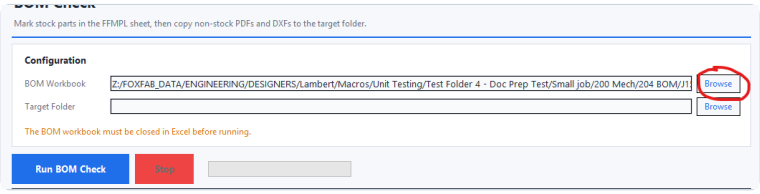
> ! *Make sure the **Everything** service is running. Open the Everything application and confirm the file count is not zero.*

4.3 Step-by-Step

Step 1. Click **Bom Filler** in the sidebar.

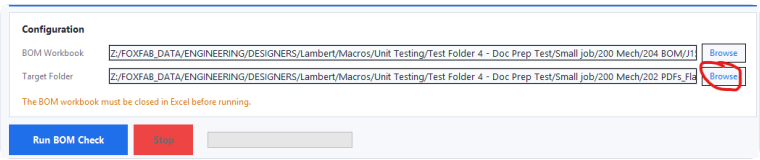


Step 2. In the **BOM Workbook** field, click **Browse** and select the job BOM `.xlsx` or `.xlsm` file. You can also type the path directly.



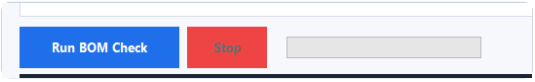
> ℹ *If you browse to a folder instead of a file, the app will list all Excel files inside it and let you pick one.*

Step 3. In the **Target Folder** field, click **Browse** and select the folder where non-stock PDFs and DXFs should be copied. This is usually the job's `202 PDFs_Flats` folder.

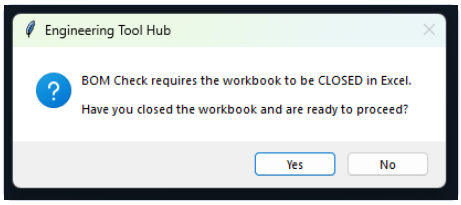


> ℹ *If the BOM is inside a `204 BOM` folder, the target folder will be **auto-populated** with the sibling `202 PDFs_Flats` path. You can accept the default or change it.*

Step 4. Close Excel (if open), then click **Run**.



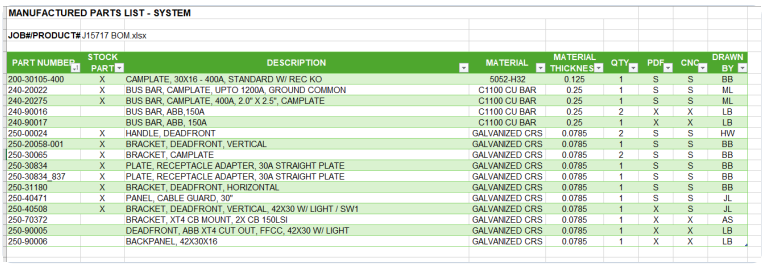
Step 5. A confirmation prompt appears asking you to verify Excel is closed. Click **Yes** to proceed.



Step 6. Watch the terminal for progress. The stock index is pre-fetched first, then Pass 1 runs, followed by Pass 2.

```
[INFO] Opening workbook:  Z:\Jobs\26-0042\204 BOM\BOM rA.xlsx
[INFO] Indexing stock folder...
[OK] Indexed 342 stock files.

============================================================
PASS 1 -- Stock Parts Check
Searching:  Z:\...300 Stock Parts\PDFs & Flats
============================================================
Checking:  300-1005      FOUND -- marked X / S / S
Checking:  240-2201      not found
Checking:  245-3010      FOUND -- marked X / S / N/A (flexibar)
Checking:  240-2202      not found

Pass 1 complete:  2 of 4 parts found in stock.

============================================================
PASS 2 -- Non-Stock PDF + DXF Copy
============================================================
240-2201  rB  PDF:COPIED  DXF:COPIED
240-2202  rC  PDF:EXISTS  DXF:EXISTS

[OK] Workbook saved.  Pass 2 complete.
```

Step 7. Once complete, the progress bar fills and a summary is shown. Open the BOM in Excel to verify the markings.



MANUFACTURED PARTS LIST - SYSTEM

JOB#/PRODUCT# J15717 BOM.xlsx

PART NUMBER	STOCK PART	DESCRIPTION	MATERIAL	MATERIAL THICKNES	QTY	PDF	CNC	DRAWN BY
200-30105-400	X	CAMPLATE, 30X16 - 400A, STANDARD W/ REC KO	5052-H32	0.125	1	S	S	BB
240-20022	X	BUS BAR, CAMPLATE, UPTO 1200A, GROUND COMMON	C1100 CU BAR	0.25	1	S	S	ML
240-20275	X	BUS BAR, CAMPLATE, 400A, 2.0" X 2.5", CAMPLATE	C1100 CU BAR	0.25	1	S	S	ML
240-90016		BUS BAR, ABB,150A	C1100 CU BAR	0.25	2	X	X	LB
240-90017		BUS BAR, ABB, 150A	C1100 CU BAR	0.25	1	X	X	LB
250-00024	X	HANDLE, DEADFRONT	GALVANIZED CRS	0.0785	2	S	S	HW
250-20058.001	X	BRACKET, DEADFRONT, VERTICAL	GALVANIZED CRS	0.0785	1	S	S	BB
250-30065	X	BRACKET, CAMPLATE	GALVANIZED CRS	0.0785	2	S	S	BB
250-30834	X	PLATE, RECEPTACLE ADAPTER, 30A STRAIGHT PLATE	GALVANIZED CRS	0.0785	1	S	S	BB
250-30834_837	X	PLATE, RECEPTACLE ADAPTER, 30A STRAIGHT PLATE	GALVANIZED CRS	0.0785	1	S	S	BB
250-31180	X	BRACKET, DEADFRONT, HORIZONTAL	GALVANIZED CRS	0.0785	1	S	S	BB
250-40471	X	PANEL, CABLE GUARD, 30"	GALVANIZED CRS	0.0785	1	S	S	JL
250-40508	X	BRACKET, DEADFRONT, VERTICAL, 42X30 W/ LIGHT / SW1	GALVANIZED CRS	0.0785	1	X	S	JL
250-70372		BRACKET, XT4 CB MOUNT, 2X CB 150LSI	GALVANIZED CRS	0.0785	1	X	X	AS
250-90005		DEADFRONT, ABB XT4 CUT OUT, FFCC, 42X30 W/ LIGHT	GALVANIZED CRS	0.0785	1	X	X	LB
250-90006		BACKPANEL, 42X30X16	GALVANIZED CRS	0.0785	1	X	X	LB

4.4 Understanding the Results

Column	Value	Meaning
B — STOCK PART	`X`	Part found in the stock library
G — PDF	`S`	Stock part — file sourced from stock
G — PDF	`X`	Non-stock — PDF/DXF copied to target folder
H — CNC	`S`	Stock part — CNC sourced from stock
H — CNC	`N/A`	Flexibar part — no DXF file exists

4.4.1 Combined-Part Files

When a combined-part file is found (e.g. `240-90123_124_125 rC.pdf`), Bom Filler automatically marks *all* covered part rows in the BOM. The terminal will show:

```
240-90123  rC  PDF:COPIED  DXF:COPIED
Combined:  also marked 240-90124 (row 42)
Combined:  also marked 240-90125 (row 43)
```

4.4.2 Punch Program Filter

Files located under `PUNCH PROGRAMS` directories are automatically excluded from search results. This ensures the tool always grabs the clean engineering copy of a PDF or DXF rather than a punch-specific variant.

4.5 Stopping Early

Click **Stop** at any time to halt the run after the current part finishes. The workbook will not be saved if stopped mid-pass — re-run from the beginning.

5 Doc Prep & Print

Doc Prep & Print builds a complete manufacturing packet from a job folder and sends it to the printer in the correct order, or saves it locally in Simulation Mode.

5.1 Overview — What Gets Printed

#	Document	Duplex	Notes
01	Fabrication Work Order	Simplex	Auto-filled from PRF data
02	BOM (all sheets)	Simplex	CNC column H already marked
03–N	CNC files (duplex)	Duplex	One PDF per file
N+1	CNC Simplex (merged)	Simplex	All simplex CNCs merged
N+2	PDFs_Flats (merged)	Simplex	All flat PDFs merged
N+3	Production Release Form	Simplex	First sheet only
N+4	Electrical Pack	Simplex	Pages 1–2 only
N+5	Assemblies (merged)	Simplex	All assembly PDFs merged

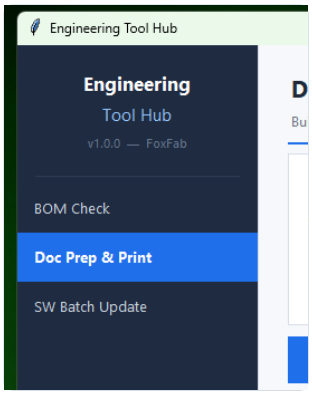
5.2 Before You Start

! *Close the BOM Excel file before clicking* ***Generate Documents****. The CNC column marking step opens the BOM via Excel — it will fail if the file is already open.*

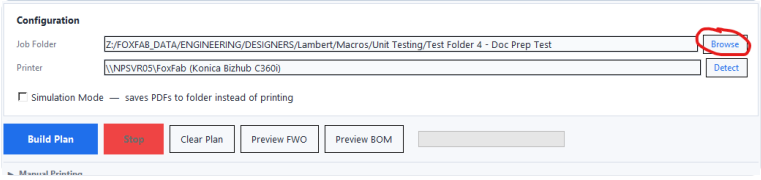
! *Adobe Acrobat (not Reader) must be installed for COM printing. If it is not found, the app will fall back to a slower printing method.*

5.3 Step-by-Step

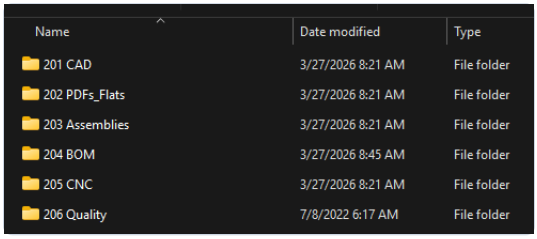
Step 1. Click **Doc Prep & Print** in the sidebar.



Step 2. In the **Job Folder** field, click **Browse** and select the job's **variant folder** — the one that contains `204 BOM\`, `205 CNC\`, and `202 PDFs_Flats\` inside it.



> *The variant folder is usually named something like `J26-0042-01` or `01 Standard`. Select the folder* containing *the `204 BOM` folder, not the `204 BOM` folder itself.*



Step 3. Check that the **Printer** field shows the correct printer name. The default is the FoxFab Konica Bizhub. Change it only if printing elsewhere.

Step 4. Click **Build Plan**.

Step 5. The terminal scans the job folder and lists everything it found. Review the output to confirm all expected documents are detected.

```
[INFO] Building plan for:  Z:\Jobs\26-0042\01 Standard
[OK] BOM: BOM rA.xlsx
[OK] PRF: PRF J26-0042.xlsx
[OK] FWO template:  FWO_Template.pdf
[OK] CNC duplex:  3 files
[OK] CNC simplex:  5 files
[OK] PDFs_Flats:  8 files
[WARN] Electrical pack not found -- will be skipped
[OK] Plan ready.  Click Generate Documents to continue.
```

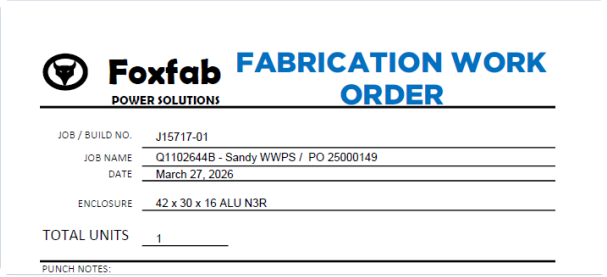
> *Items shown as `[WARN]` will be skipped — the run does not stop on a missing document.*

5.4 Preview Options (Optional)

Before committing to a full print run, use the preview buttons to verify key documents look correct.

5.4.1 Preview FWO

Click **Preview FWO** to fill and open the Fabrication Work Order without printing it. The result opens automatically from `logs\FWO_preview.pdf`.



Foxfab POWER SOLUTIONS

FABRICATION WORK ORDER

JOB / BUILD NO. J15717-01

JOB NAME Q1102644B - Sandy WWPS / PO 25000149

DATE March 27, 2026

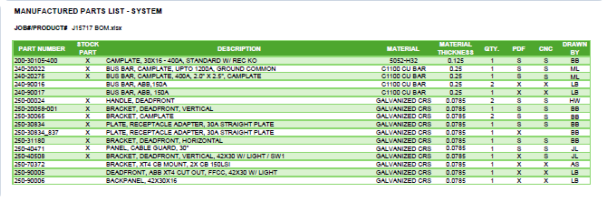
ENCLOSURE 42 x 30 x 16 ALU N3R

TOTAL UNITS 1

PUNCH NOTES:

5.4.2 Preview BOM

Click **Preview BOM** to run the CNC marking pass first, then export and open the BOM as a PDF. Use this to confirm column H is correct before printing.



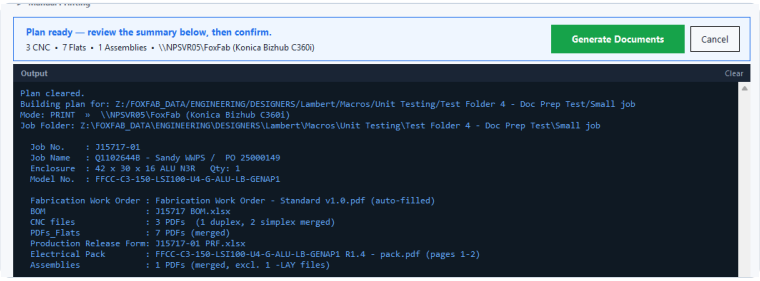
MANUFACTURED PARTS LIST - SYSTEM

JOB#/PRODUCT# J15717 BOM.xlsx

PART NUMBER	STOCK PART	DESCRIPTION	MATERIAL	MATERIAL THICKNESS	QTY.	PDF	CNC	DRAWN BY
200-30105-400	X	CAMPLATE, 30X16 - 400A, STANDARD W/ REC KO	5052-H32	0.125	1	S	S	BB
240-20022	X	BUS BAR, CAMPLATE, UPTO 1200A, GROUND COMMON	C1100 CU BAR	0.25	1	S	S	ML
240-20278	X	BUS BAR, CAMPLATE, 400A, 2.0" X 2.5", CAMPLATE	C1100 CU BAR	0.25	1	S	S	ML
240-90116		BUS BAR, ABB, 160A	C1100 CU BAR	0.25	2	X	X	LB
240-90117		BUS BAR, ABB, 160A	C1100 CU BAR	0.25	1	X	X	LB
250-00024	X	HANDLE, DEADFRONT	GALVANIZED CRS	0.0795	2	S	S	HW
250-20066-001	X	BRACKET, DEADFRONT, VERTICAL	GALVANIZED CRS	0.0795	1	S	S	BB
250-30064	X	BRACKET, CAMPLATE	GALVANIZED CRS	0.0795	2	S	S	BB
250-30834	X	PLATE, RECEPTACLE ADAPTER, 30A STRAIGHT PLATE	GALVANIZED CRS	0.0795	1	S	S	BB
250-30834_837	X	PLATE, RECEPTACLE ADAPTER, 30A STRAIGHT PLATE	GALVANIZED CRS	0.0795	1	X		BB
250-31180	X	BRACKET, DEADFRONT, HORIZONTAL	GALVANIZED CRS	0.0795	1	S	S	BB
250-40471	X	PANEL, CABLE GUARD, 30"	GALVANIZED CRS	0.0795	1	S	S	JL
250-40908	X	BRACKET, DEADFRONT, VERTICAL, 42X30 W/ LIGHT / BW1	GALVANIZED CRS	0.0795	1	X	S	JL
250-70372		BRACKET, XT4 CB MOUNT, 24 CB 160LSI	GALVANIZED CRS	0.0795	1	X	X	AG
250-90005		DEADFRONT, ABB XT4 CUT OUT, FFCC, 42X30 W/ LIGHT	GALVANIZED CRS	0.0795	1	X	X	LB
250-90006		BACKPANEL, 42X30X16	GALVANIZED CRS	0.0795	1	X	X	LB

> ! *Preview BOM opens the BOM via Excel COM. Close Excel before clicking it.*

> **Step 6.** Once the plan looks correct, click **Generate Documents**.



The app will automatically:

a. Scan `205 CNC\` and mark column H of the BOM with `X` for every matched part, then save the workbook.

b. Generate all documents into a temporary folder in print order.

c. Auto-populate the Manual Printing list with every generated PDF.

```
[INFO] CNC Column Marker:  scanning 205 CNC...
[OK] Found 8 CNC PDFs in 205 CNC
[OK] Resolved 14 unique part numbers
[OK] BOM marked and saved.
[INFO] Generating 01_FWO...
[OK] 01_FWO_J26-0042.pdf
[INFO] Generating 02_BOM...
[OK] 02_BOM_rA.pdf
...
[OK] All documents generated.  Ready to print.
```

Step 7. An **amber confirmation bar** appears listing every document queued for printing. Read through the list before proceeding.

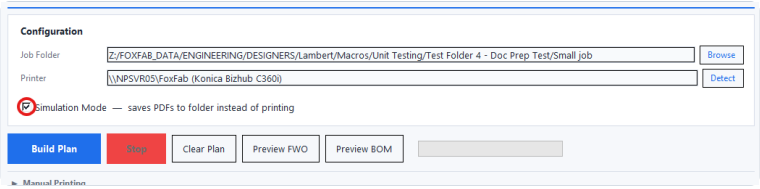
Step 8. Click **Send to Printer** when ready. All documents are sent to Acrobat in order. Do not close the app while printing is in progress.

```
[INFO] Starting Acrobat COM session...
[OK] Printing 01_FWO_J26-0042.pdf   (simplex)
[OK] Printing 02_BOM_rA.pdf       (simplex)
[OK] Printing 03_CNC_J26042-01.pdf  (duplex)
...
[OK] All documents sent.  Acrobat session closed.
```

ℹ *Clicking **Cancel Print** instead sends nothing to the printer and clears the queue. You can adjust anything and click Generate Documents again.*

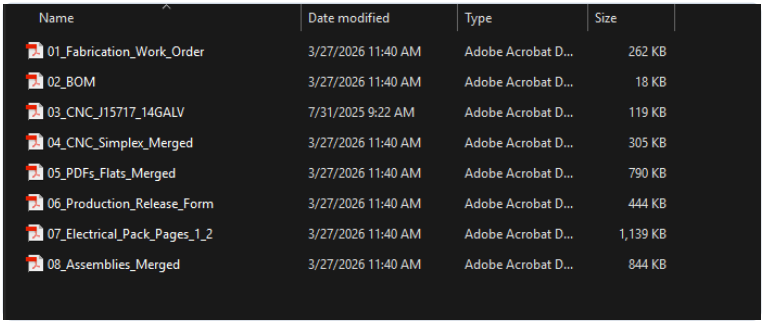
5.5 Simulation Mode

Tick the **Simulation Mode** checkbox before clicking **Send to Printer** to save all documents locally instead of printing. This is useful for reviewing output on a machine without a printer.



Documents are saved to a timestamped folder next to the application:

`Simulated_Print_Output_J26-0042_20260327_091500\`



Name	Date modified	Type	Size
01_Fabrication_Work_Order	3/27/2026 11:40 AM	Adobe Acrobat D...	262 KB
02_BOM	3/27/2026 11:40 AM	Adobe Acrobat D...	18 KB
03_CNC_J15717_14GALV	7/31/2025 9:22 AM	Adobe Acrobat D...	119 KB
04_CNC_Simplex_Merged	3/27/2026 11:40 AM	Adobe Acrobat D...	305 KB
05_PDFs_Flats_Merged	3/27/2026 11:40 AM	Adobe Acrobat D...	790 KB
06_Production_Release_Form	3/27/2026 11:40 AM	Adobe Acrobat D...	444 KB
07_Electrical_Pack_Pages_1_2	3/27/2026 11:40 AM	Adobe Acrobat D...	1,139 KB
08_Assemblies_Merged	3/27/2026 11:40 AM	Adobe Acrobat D...	844 KB

5.6 Manual Printing

At the bottom of the Doc Prep panel is a collapsible » **Manual Printing** section. Click the arrow to expand it.

- After **Generate Documents** the list is **auto-populated** with every generated PDF in print order, duplex pre-set per file.
- Click **Add File** to add any extra PDF manually.
- Click **X** next to a file to remove it from the list.
- **Print** sends the selected file immediately (background thread — UI stays responsive).
- **Print All** sends all files in list order through the same Acrobat session.

Use Manual Printing to resend a single document or the full batch without re-running the entire plan.

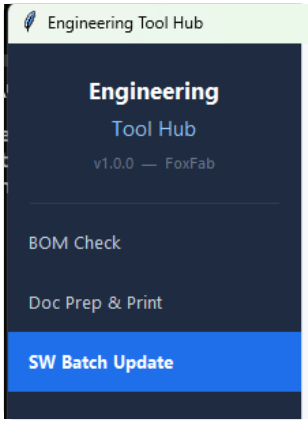
6 SW Batch Update

SW Batch Update guides you through running the SolidWorks VB macro that batch-updates custom properties and exports DXF flat patterns.

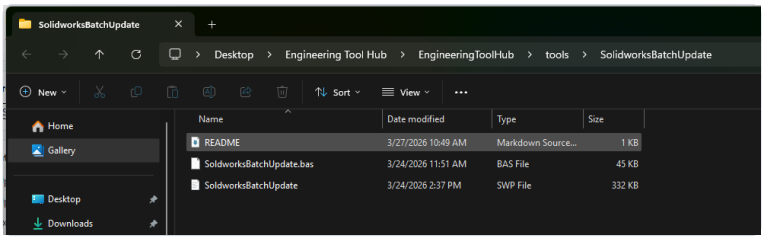
> ! *SolidWorks must be installed on your machine. This tool cannot function without it.*

6.1 Step-by-Step

Step 1. Click **SW Batch Update** in the sidebar.

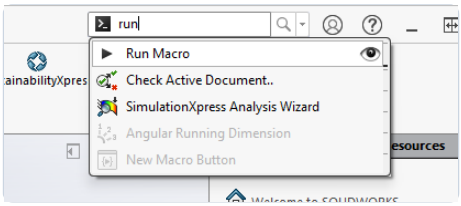


Step 2. The panel shows the path to the macro file. Click **Open Folder** to open the containing folder in File Explorer.



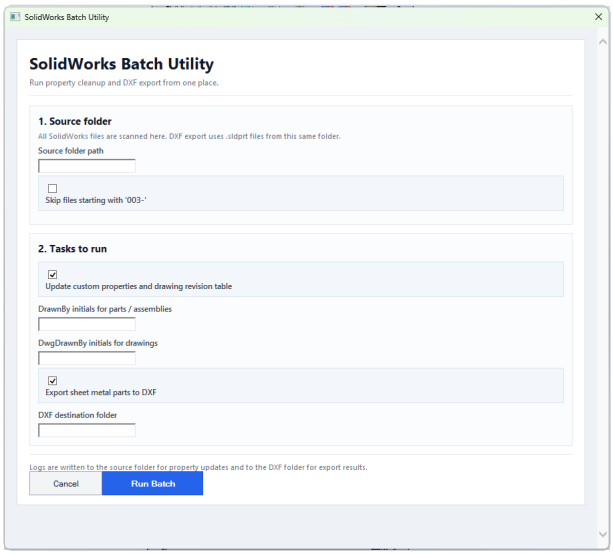
Step 3. Open SolidWorks.

Step 4. In SolidWorks, go to **Tools** → **Macro** → **Run**.



Step 5. Browse to `SoldworksBatchUpdate.bas` and click **Open**.

Step 6. The macro dialog opens. Enter your initials, select the source folder and output folder, then click **Run**.



Step 7. The macro updates `DrawnBy` and `DwgDrawnBy` properties across all parts in the folder and exports flat-pattern DXF files.

7 SW Batch PDF Export

SW Batch PDF Export provides a step-by-step guide for using SolidWorks Task Scheduler to batch-export drawings as PDF files. This tab is purely instructional — it does not execute anything automatically.

7.1 Step-by-Step

Step 1. Click **SW Batch PDF Export** in the sidebar.

Step 2. Open **SolidWorks Task Scheduler** from the Windows Start menu or from within SolidWorks via **Tools** → **Task Scheduler**.

Step 3. Select **Convert Files** from the Task Scheduler menu.

Step 4. Set the **Source** to the folder containing the SolidWorks drawing files you want to export.

Step 5. Set the **Output Format** to `PDF`.

Step 6. Choose the **Destination** folder where the exported PDFs should be saved.

Step 7. Click **Run Now** to start the batch export.

7.2 Tips

- Use **folder mode** to process an entire directory of drawings at once.
- Enable **sub-folder recursion** if drawings are nested in subdirectories.
- Paper size behaviour is determined by the drawing template — Task Scheduler does not override it.

8 File Logger (Parts Tracker)

File Logger launches the standalone **Parts Tracker** application, a PyQt6 desktop tool for tracking SolidWorks part numbers across job folders.

8.1 Features

- **Part number tracking** — Tracks part numbers across all job folders on the network drive.
- **Gap-aware suggestions** — Suggests the next available part number, accounting for gaps in the existing sequence.
- **Orphan detection** — Identifies part files that are not referenced in any job BOM.
- **Archive classification** — Classifies parts as active or archived based on job status.
- **Live filesystem watcher** — Monitors for new or changed part files in real time.

8.2 Launching

Step 1. Click **File Logger** in the sidebar.

Step 2. Click the **Launch** button. The Parts Tracker application opens in a separate window.

i *File Logger requires Python 3.10+, PyQt6, openpyxl, and requests to be installed. It also requires the Everything HTTP Server to be running on port 8080.*

! *The Parts Tracker runs as a separate process. Closing Engineering Tool Hub does not automatically close the Parts Tracker window.*

9 Training Materials

The Training Materials tab provides quick access to engineering design reference documents and design tips.

9.1 Engineering Design Package

This section lists all reference documents stored in the `tools\EngineeringDesignPackage\` folder. Click **Print All** to send every document to the printer double-sided in a single background thread.

The following documents are included:

#	Document
1	CU cut tool.pdf
2	FLEXI HOW TO CONNECT.pdf
3	Flexibar Advanced Technical Characteristics.pdf
4	Minimum Bending Space.pdf
5	Punch tooling list.pdf
6	UL891 Annex G.pdf
7	Wire Connections — Internal.pdf

- The status label updates in real time during printing (e.g. "Printing 3/7: UL891 Annex G.pdf…").
- The print button is disabled while printing is in progress and re-enables when the job completes.
- All documents are printed **duplex** (double-sided) using the same Acrobat COM infrastructure as Doc Prep & Print.

9.2 Design Reference

Click **Open FoxFab Design Tips** to open the `FoxFab_Design_Tips.docx` document in your default application (typically Microsoft Word). This provides quick access to FoxFab design guidelines without leaving the tool.

10 Troubleshooting

If something isn't working as expected, check the table below for the most common issues and their solutions. The terminal output usually contains a clear error message — read it first before looking here.

Symptom	Solution
App will not start	Ensure the entire `Engineering Tool Hub\` folder is present, not just the `.exe`. If files are missing from `_internal\`, re-run `build.bat` and redistribute the whole folder.
`es.exe not found`	Everything CLI is missing. Check that `es.exe` is in the same folder as the `.exe` file.
"Could not open workbook"	The BOM file is still open in Excel. Close it fully (check the taskbar) and run again.
Pass 1 marks everything as "not found"	The Everything service is not running or has not indexed `Z:\`. Open Everything, go to **Tools → Rebuild Database**, and wait for indexing to complete.
Flexibar parts show "not found" for DXF	This should not happen in v1.2.0. Parts with the `245-` prefix are automatically marked `N/A`. Verify you are running v1.2.0+ (check the sidebar version label).
CNC folder not found	The app looks for a folder beginning with `205` inside the variant folder. If named differently, a picker opens — browse to it and click **Select**. Cancel to skip CNC marking.
FWO text in wrong position	In `app.py`, find the `FWO_*` constants near the top and adjust the `X/Y` values. Use **Preview FWO** to verify.
Acrobat not found	Adobe Acrobat (not Reader) must be installed. For non-standard install paths, add the full path to `ACROBAT_SEARCH_PATHS` in `app.py`.
Printing stops partway	Use the **Manual Printing** panel to resend documents that did not print. Add files individually and click **Print** or **Print All**.
File Logger will not launch	Ensure Python 3.10+ is on the system PATH. Install missing dependencies via `pip install PyQt6 openpyxl requests`.
How to Use shows "PDF not found"	Place `Engineering_Tool_Hub.pdf` next to the `.exe` or re-run `build.bat` to copy it into the distribution folder.

11 Quick Reference

11.1 Sidebar Navigation

#	Tab	Purpose
1	How to Use	Embedded PDF user guide (default landing)
2	Bom Filler	Stock check + non-stock file copy
3	Doc Prep & Print	Manufacturing packet builder and printer
4	SW Batch Update	SolidWorks macro launcher
5	SW Batch PDF Export	Task Scheduler PDF export guide
6	File Logger	Parts Tracker application launcher
7	Training Materials	Reference document printing + design tips

11.2 Button Reference

Button / Element	Purpose
Sidebar tool name	Switch to that tool panel instantly
Build Plan	Scan the job folder and report what was found
Generate Documents	Mark CNC column, generate all PDFs in print order
Run	Start the Bom Filler workflow
Stop	Halt the running task after the current step
Browse	Open a file or folder picker
Preview FWO	Fill and open the FWO PDF without printing
Preview BOM	Mark CNC column and open BOM PDF without printing
Send to Printer	Commit the document queue to the printer
Launch	Open the Parts Tracker (File Logger tab)
Print All	Print all training materials duplex
Open FoxFab Design Tips	Open design tips document in Word
Prev / **Next**	Navigate pages in the How to Use viewer
Simulation Mode checkbox	Save output locally instead of printing
» Manual Printing	Expand/collapse the manual print queue

11.3 BOM Column Marking Reference

Column	Value	Meaning
B — STOCK PART	`X`	Found in stock library
G — PDF	`S`	Stock — file from stock folder
G — PDF	`X`	Non-stock — PDF/DXF copied to target
H — CNC	`S`	Stock — CNC from stock folder
H — CNC	`N/A`	Flexibar part (245- prefix) — no DXF exists
H — CNC	`X`	CNC file found and matched in 205 CNC

11.4 Print Document Order

#	Document	Duplex	Pages
01	Fabrication Work Order	Simplex	All
02	BOM (all sheets)	Simplex	All
03–N	CNC duplex files	Duplex	All
N+1	CNC simplex (merged)	Simplex	All
N+2	PDFs_Flats (merged)	Simplex	All
N+3	Production Release Form	Simplex	1 only
N+4	Electrical Pack	Simplex	1–2 only
N+5	Assemblies (merged)	Simplex	All

12 Frequently Asked Questions

Q: Do I need to install Python if I'm only using the `.exe`?

A: No. The packaged `.exe` includes its own Python runtime inside the `_internal` folder. A separate Python installation is only needed if you want to run the source code directly or use the File Logger / Parts Tracker.

Q: Can I move or rename the `Engineering Tool Hub` folder?

A: Yes — you can move or rename the top-level folder freely. Just keep the `.exe` and the `_internal` folder together in the same directory. Do not move the `.exe` out on its own.

Q: Does the BOM file need to be in a specific format?

A: The workbook must contain a sheet named `FFMPL`. The tool reads part numbers, revision levels, and stock indicators from specific columns on that sheet. Standard FoxFab BOM templates already include this sheet.

Q: What happens if I run Bom Filler on a BOM that was already processed?

A: Bom Filler is safe to re-run. Stock markings are overwritten with the latest results, and file copies that already exist in the target folder are skipped (shown as `EXISTS` in the terminal).

Q: Can I print to a different printer?

A: Yes. In Doc Prep & Print, change the **Printer** field to any installed printer name before clicking **Generate Documents**. Duplex settings are applied per-document regardless of printer.

Q: What is Simulation Mode?

A: Simulation Mode saves all generated documents to a timestamped local folder instead of sending them to the printer. Use it to review output on a machine without a printer or to verify everything looks correct before committing to a print run.

Q: Why does the app ask me to close Excel before every run?

A: Bom Filler and Doc Prep both write to the BOM workbook via Excel COM automation. If the file is already open in another Excel instance, the COM session cannot acquire a write lock and the operation will fail.

Q: What is a "combined-part" file?

A: Some engineering drawings cover multiple part numbers in a single PDF, e.g. `240-90123_124_125 rC.pdf` covers parts 240-90123, 240-90124, and 240-90125. Bom Filler detects these filenames and cross-marks all covered rows in the BOM automatically.

Q: Do I need the Everything HTTP server for all tools?

A: Only Bom Filler and File Logger require the Everything HTTP server (port 8080). Doc Prep & Print, SW Batch Update, SW Batch PDF Export, and Training Materials work independently of Everything.

Q: How do I check which version I'm running?

A: The version number is shown in two places: the sidebar footer (bottom-left of the window) and the splash screen on startup. This guide covers **v1.2.0**.

Q: Can I add my own documents to the Training Materials print list?

A: Place additional PDF files in the `tools\EngineeringDesignPackage\` folder. They will appear in the list automatically the next time you open the Training Materials tab.

Q: The Parts Tracker window stays open after I close Engineering Tool Hub. Is that normal?

A: Yes. File Logger launches Parts Tracker as a separate process. Closing the main app does not close Parts Tracker — close it independently when you are done.